

Algebra-Based Physics-1: Mechanics (PHYS135A): Unit 0

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Opening Remarks - Welcome!

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Figure 1: Taking physics for the first time.

Summary

Week 1 Summary

Physics - φυσική - "phusiké": knowledge of nature
from φύσις - "phúsis": nature

1. Estimation/Unit Analysis - Chapters 1.1 - 1.4
 - **Estimating** the correct order of magnitude
 - **Unit analysis** - dealing with the units of numbers
2. Coordinates and vectors - Chapters 2.1 - 2.3
 - **Scalars** and **vectors**
 - **Cartesian** (rectangular) coordinates, displacement
 - **Vector** addition, subtraction, and multiplication
3. Review of Geometry and Trigonometry Techniques
 - Types of **triangles**, special angles
 - $\sin(x)$, $\cos(x)$, $\tan(x)$
 - Pythagorean theorem

Estimation/Unit Analysis - Chapters 1.1 - 1.4

Estimation/Unit Analysis

In science and engineering, **estimation** is to obtain a quantity in the absence of precision, informed by rational constraints.

1. Define relevant **unit scales**: (mg, g, or kg), (m/s or km/hr)
2. Obtain **complex quantities** from simple ones
 - Obtain *areas* and *volumes* from *lengths*
 - Obtain *rates* from *numerators* and *denominators*
3. Taking advantage of **scaling problems**
 - Knowing *relationship* between variables
 - Using that *relationship* to obtain new information
4. Constrain the unknown with **upper** and **lower** limits

Units: Which of the following represents a *volume*?

- A: 10 gm
- B: 10 cm²
- C: 1 cm³
- D: 1 cm s⁻¹

Estimation/Unit Analysis

Units: If a grain of sand within a fluid sinks 15 cm in 5 seconds, what is the speed of the grain?

- A: 3 cm
- B: 3 s
- C: 3 s/cm
- D: 3 cm/s

Unit conversion: If a person weights 120 lbs, what is their weight in kilograms?¹

- A: 54.5 kg
- B: 264 kg
- C: 54.5 lbs
- D: 264 lbs

¹One kilogram is 2.2 lbs.

Unit conversion: A **density** is a mass divided by a volume. For example, water has a density of 1 gm cm^{-3} . What is the density of water in kg m^{-3} ?

- A: 1 kg m^{-3}
- B: 10 kg m^{-3}
- C: 100 kg m^{-3}
- D: 1000 kg m^{-3}

Group exercise on complex units: A *vitrolero* is a classic container for serving *agua fresca*. It has a diameter of 20 cm, and a height of 30 cm. How many cups can we serve from the vitrolero if we put 0.5 liters of agua fresca in each cup?

- *Hint:* 1 liter is 1000 mL
- *Hint:* 1 mL is 1 cm³
- **Volume:** The volume of a cylinder is π times the radius of the base, squared, , times the height: $\pi r^2 h$.

Estimation/Unit Analysis

Unit scale: A generation is about one-third of a lifetime.
Determine how many generations have passed since the year 0 AD².

- A: 10
- B: 20
- C: 60
- D: 100

²What is the appropriate scale here?

Estimation/Unit Analysis

Unit scale: (a) What fraction of Earth's diameter³ is the greatest ocean depth (11 km below sea level)? (b) The greatest mountain height (8.8 km above sea level)?

- A: 8.6×10^{-2} , 6.9×10^{-2}
- B: 8.6×10^{-3} , 6.9×10^{-3}
- C: 8.6×10^{-4} , 6.9×10^{-4}
- D: 8.6×10^{-5} , 6.9×10^{-3}

³The diameter of the Earth is 12,800 km.

Complex quantities: If a Whittier College athlete ran the 5k race at a track meet in 35 minutes, what was her average speed?

- A: 0.3 meters per second
- B: 3 meters per second
- C: 30 meters per second
- D: 300 meters per second

Complex quantities: Suppose you won the lottery and received \$1 billion USD. Because your life is dope, you stack that paper over the Whittier College soccer field. Each stack contains 100 bills, and each bill is worth \$100. If you evenly cover the field, how high is the money level?

- A: 0.5 inch
- B: 1 inch
- C: 2 inches
- D: 1 foot

Scaling problem: Supposed you have an ideal gas in a cylinder of fixed volume. If the pressure begins as 100 kPa, and you *double* the temperature of the gas, what is the new pressure?

- A: 100 kPa
- B: 50 kPa
- C: 10 kPa
- D: 200 kPa

Scaling problem: Supposed you have an ideal gas in a cylinder of fixed volume. If the pressure begins as 100 kPa, and you *halve* the temperature of the gas, what is the new pressure?

- A: 100 kPa
- B: 50 kPa
- C: 10 kPa
- D: 200 kPa

Upper/lower limits: How many undergraduate students are there at Whittier College⁴?

- A: 5,000
- B: 1,000
- C: 1,250
- D: 500

⁴What is the absolute lower limit, and what is the upper limit?

Estimation/Unit Analysis

Upper/lower limits: What is the average yearly college tuition in the United States (before subtracting grants and scholarships)?

- A: \$5,000
- B: \$10,000
- C: \$25,000
- D: \$40,000

What information affects the **upper** and **lower** limits here?

Coordinates and Vectors - Chapters 2.1 - 2.3

Activity Link

Who understands coordinates and vectors better than anyone else?

https://youtu.be/0B7WL7nhIF4?si=_dl4t_GwL98aXWFa

Coordinates and Vectors - Scalars, Vectors (Chapters 2.1-2.3)

Physics requires **mathematical objects** to build equations that capture the behavior of nature. Two examples of such objects are **scalar** and **vector** quantities. Each type of object obeys similar but different rules.

1. Scalar quantities

- mass: $m_1 + (m_2 + m_3) = (m_1 + m_2) + m_3$
- speed: $v_1(v_2 + v_3) = v_1v_2 + v_1v_3$
- charge: $q_1\left(\frac{1}{q_1}\right) = 1, q_1(0) = 0$

2. Vector quantities

- velocity: $\vec{v}_1 + (\vec{v}_2 + \vec{v}_3) = (\vec{v}_1 + \vec{v}_2) + \vec{v}_3$
- tension: $\vec{t}_1 \cdot (\vec{t}_2 + \vec{t}_3) = \vec{t}_1 \cdot \vec{t}_2 + \vec{t}_1 \cdot \vec{t}_3$

Coordinates and Vectors - Scalars, Vectors (Chapters 2.1-2.3)

A vector may be expressed as *a list of scalars*: $\vec{v} = (4, 2)$ (a vector with two *components*), $\vec{u} = (3, 4, 5)$ (three *components*). Now, we know how to add and subtract scalars. How do we add and subtract vectors?

What is

$(1, 3, 8) +$

$(0, 2, 1)$?

Answer: $(1, 5, 9)$

In other words, when adding vectors, we add them component by component.

Coordinates and Vectors - Scalars, Vectors (Chapters 2.1-2.3)

How do we subtract vectors? In the same fashion:

What is

$(1, 3, 8) -$

$(0, 2, 1)$?

Answer: $(1, 1, 7)$

In other words, when subtracting vectors, we subtract them component by component.

Coordinates and Vectors - Scalars, Vectors (Chapters 2.1-2.3)

A HTML-based demonstration for adding vectors:

`https://phet.colorado.edu/en/simulations/
vector-addition`

Notice several things:

- Produce vectors that *cancel* each other.
- What happens when vectors are parallel and orthogonal?

Coordinates and Vectors - Scalars, Vectors (Chapters 2.1-2.3)

How do we multiply vectors? In the same fashion, *for one kind of multiplication*:

What is

$$(1, 3, 8) \cdot (0, 2, 1)?$$

$$\text{Answer: } 1 \cdot 0 + 3 \cdot 2 + 8 \cdot 1 = 14$$

This kind of multiplication is known as the dot-product. There is also the *cross-product*, which we will save for later.

Coordinates and Vectors - Coordinates (Chapters 2.1-2.3)

The components of a vector may describe quantities in a **coordinate system**, such as *Cartesian coordinates* - after René Descartes.

Vectors in the 3D Cartesian coordinate system (x,y,z) may be written in the following notation:

$$\vec{v} = a\hat{i} + b\hat{j} + c\hat{k}$$

- a: The amount in the +x-direction, $\hat{i}$: a vector of length 1, in the +x-direction
- b: The amount in the +y-direction, $\hat{j}$: a vector of length 1, in the +y-direction
- c: The amount in the +z-direction, $\hat{k}$: a vector of length 1, in the +z-direction

Coordinates and Vectors - Vectors (Chapters 2.1-2.3)

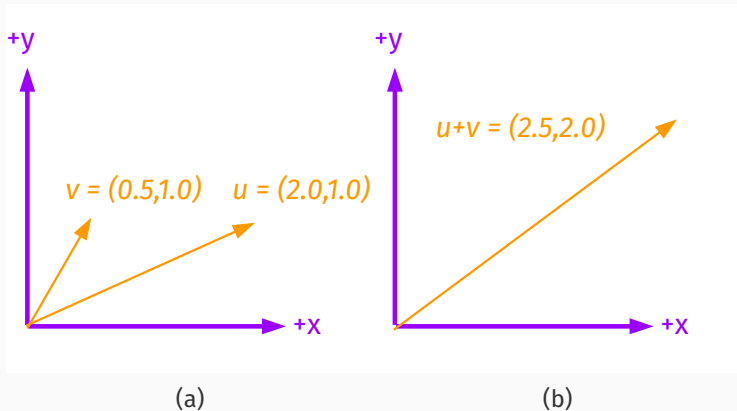


Figure 2: (a) Two vectors in a two-dimensional Cartesian coordinate system: $\vec{u} = 0.5\hat{i} + 1.0\hat{j}$ and $\vec{v} = 2.0\hat{i} + 1.0\hat{j}$. (b) What is $\vec{u} + \vec{v}$? Adding components: $\vec{u} + \vec{v} = 2.5\hat{i} + 2.0\hat{j}$.

Coordinates and Vectors - Vectors (Chapters 2.1-2.3)

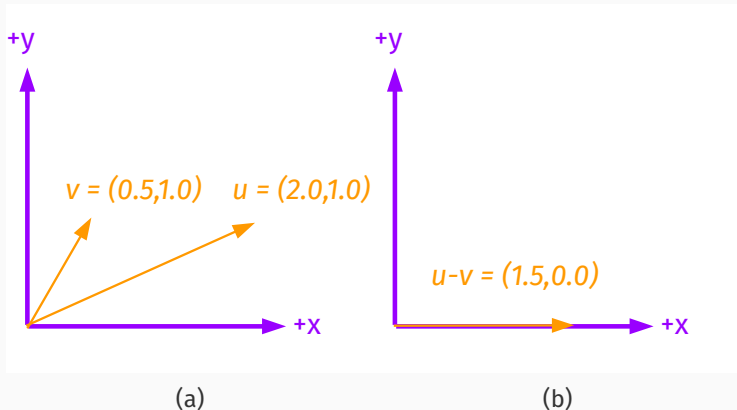


Figure 3: (a) Two vectors in a two-dimensional Cartesian coordinate system: $\vec{u} = 0.5\hat{i} + 1.0\hat{j}$ and $\vec{v} = 2.0\hat{i} + 1.0\hat{j}$. (b) What is $\vec{u} - \vec{v}$? Subtracting components: $\vec{u} - \vec{v} = 1.5\hat{i} + 0.0\hat{j}$.

Coordinates and Vectors - Vectors (Chapters 2.1-2.3)

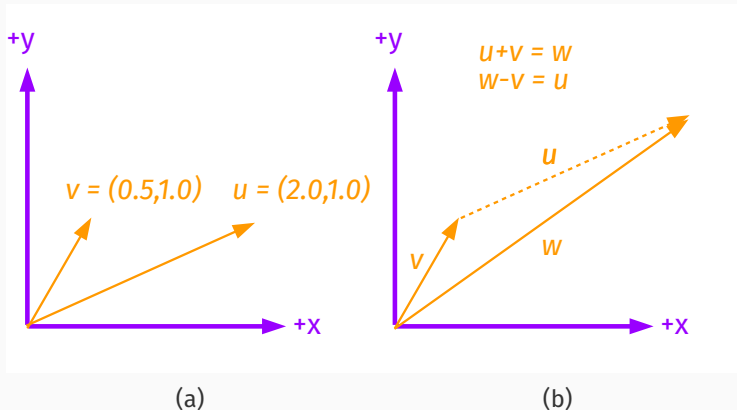


Figure 4: (a) Two vectors in a two-dimensional Cartesian coordinate system: $\vec{u} = 0.5\hat{i} + 1.0\hat{j}$ and $\vec{v} = 2.0\hat{i} + 1.0\hat{j}$. (b) To compute $\vec{w} - \vec{v}$, arrange the vectors to get a sense of the result, $\vec{u}$.

Coordinates and Vectors - Vectors (Chapters 2.1-2.3)

$$\vec{p} = 4\hat{i} + 2\hat{j}. \quad \vec{q} = -4\hat{i} + 2\hat{j}.$$

Compute $\vec{p} \cdot \vec{q}$.

- A: 12
- B: -12
- C: 4
- D: 8

$$\vec{p} = -1\hat{i} + 6\hat{j}. \quad \vec{q} = 3\hat{i} + 0.5\hat{j}.$$

Compute $\vec{p} \cdot \vec{q}$.

- A: -1
- B: 1
- C: 0
- D: 3

Coordinates and Vectors - Vectors (Chapters 2.1-2.3)

Why was the last answer zero? Look at it graphically:

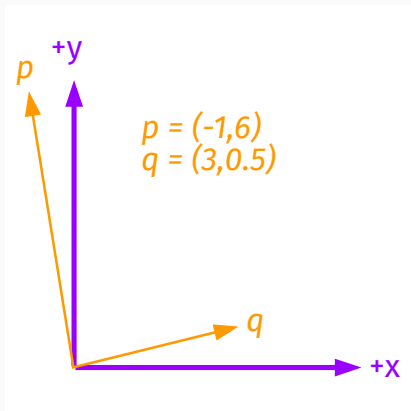


Figure 5: Two vectors $\vec{p}$ and $\vec{q}$ are *orthogonal* if $\vec{p} \cdot \vec{q} = 0$.

Coordinates and Vectors - Vectors (Chapters 2.1-2.3)

What if the vectors are parallel? Look at it graphically:

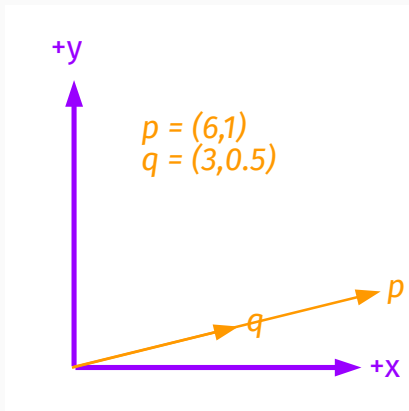


Figure 6: Two vectors $\vec{p}$ and $\vec{q}$ are *parallel* if $\vec{p} \cdot \vec{q}$ is maximal.

Coordinates and Vectors - Dot Product (Chapters 2.1-2.3)

The *length* or *norm* of a vector $\vec{v} = a\hat{i} + b\hat{j}$ is $|\vec{v}| = \sqrt{a^2 + b^2}$.

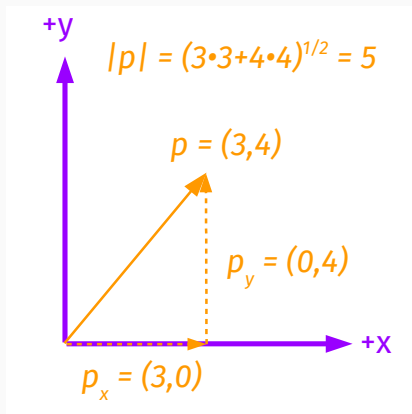


Figure 7: Computing the norm of a vector $\vec{p}$.

Coordinates and Vectors - Dot Product (Chapters 2.1-2.3)

Notice that $\sqrt{\vec{p} \cdot \vec{p}} = |\vec{p}|$.

Let θ_p be the angle between $\vec{p}$ and the x-axis.

$$p_x = \vec{p} \cdot \hat{i} = |\vec{p}| \cos(\theta_p).$$

$$p_y = \vec{p} \cdot \hat{j} = |\vec{p}| \sin(\theta_p).$$

Theorem: The dot product of two vectors $\vec{p}$ and $\vec{q}$ is $|\vec{p}||\vec{q}| \cos(\theta)$, if θ is the angle between them.

$$\begin{aligned} \text{Proof: } \vec{p} \cdot \vec{q} &= p_x q_x + p_y q_y = |p||q| \cos \theta_p \cos \theta_q + |p||q| \sin \theta_p \sin \theta_q \\ &= |p||q| (\cos \theta_p \cos \theta_q + \sin \theta_p \sin \theta_q) = |p||q| \cos(\theta_p - \theta_q) \\ &= |p||q| \cos \theta. \end{aligned}$$

$$\boxed{\vec{p} \cdot \vec{q} = |p||q| \cos \theta}$$

Coordinates and Vectors - Dot Product (Chapters 2.1-2.3)

An object moves at 2 m/s at $\theta = 60^\circ$ with respect to the x-axis. What is the velocity of the object?

- A: $(1\hat{i} + 1\hat{j})$ m/s
- B: $(\sqrt{3}\hat{i} + 1\hat{j})$ m/s
- C: $(\sqrt{3}\hat{i} + \sqrt{3}\hat{j})$ m/s
- D: $(1\hat{i} + \sqrt{3}\hat{j})$ m/s

What is the dot product of this velocity with another velocity: 5 m/s along the x-axis?

- A: 1 (m/s)^2
- B: 5 (m/s)^2
- C: 10 (m/s)^2
- D: 5 (m/s)

Coordinates and Vectors - Scalars, Vectors (Chapters 2.1-2.3)

Is it possible to multiply vectors and scalars? Of course:

$$a_1\vec{p} = a_1p_x\hat{i} + a_1p_y\hat{j}.$$

Also, multiplication properties still hold. For example:

$$(a_1 + a_2)\vec{p} = a_1\vec{p} + a_2\vec{p}.$$

A spacecraft moves at 400 m/s, at an angle of 30 degrees with respect to the x-axis. If it fires two thrusters that boost the x-component and y-component of the velocity by 25% and 50%, respectively, what is the final velocity?

- A: $(433\hat{i} + 300\hat{j})$ m/s
- B: $(300\hat{i} + 433\hat{j})$ m/s
- C: 400 m/s
- D: $(400\hat{i} + 433\hat{j})$ m/s

Coordinates and Vectors - Displacement (Chapters 2.1-2.3)

We define the *position* of an object as a vector locating it in a given coordinate system. The scalar *distance* is the norm of the position vector, that is, the distance to to the origin.

Now we can introduce the concept of **displacement**: a vector describing a movement of an object.

Coordinates and Vectors - Displacement (Chapters 2.1-2.3)

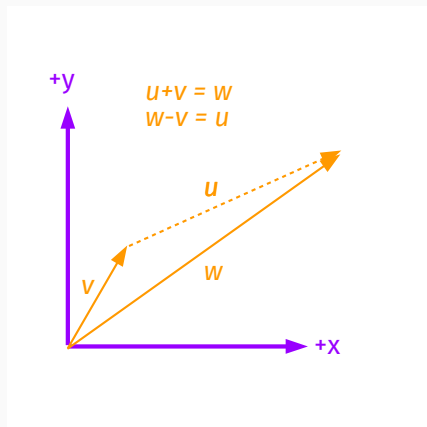


Figure 8: Suppose an object moves from position $\vec{v}$ to $\vec{w}$. In this case, the displacement is $\vec{u}$. Thus, the final position is the initial position, plus the displacement.

Coordinates and Vectors - Displacement (Chapters 2.1-2.3)

It follows that the *displacement* is zero if the initial and final positions are the same, but the *distance travelled* is not.

Suppose a jet fighter travelling at 800 km per hour banks such that it flies in a circle of radius 0.5 km. How long does it take to complete the circle? What is the distance traveled, and what is the displacement?

- A: 2π km, 28 seconds, 2π km
- B: π km, 14 seconds, π km
- C: π km, 28 seconds, π km
- D: π km, 14 seconds, 0 km

Coordinates and Vectors - Average Velocity (Chapter 3.1)

Average velocity is the ratio of the **displacement** to the elapsed time.

$$\boxed{\vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t}} \quad (1)$$

The *average speed* is the norm of the average velocity:

$$\boxed{v_{\text{avg}} = \frac{|\Delta \vec{x}|}{\Delta t}} \quad (2)$$

If the motion is in one dimension, then the average speed is

$$v_{\text{avg}} = \frac{x_f - x_i}{t_f - t_i} \quad (3)$$

Coordinates and Vectors - Average Velocity (Chapter 3.1)

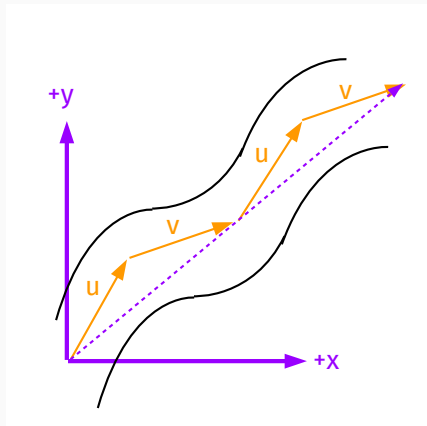


Figure 9: A Formula-1 driver keeps his car on the track by following a path approximated by the position vectors u , v , u , and v . The dashed arrow represents the total displacement.

Coordinates and Vectors - Average Velocity (Chapter 3.1)

If $\vec{u} = (20\hat{i} + 30\hat{j})$ m, and $\vec{v} = (30\hat{i} + 20\hat{j})$ m, what is the total displacement? If the elapsed time is 10 seconds, what is the average velocity?

- A: $(50\hat{i} + 50\hat{j})$ m, 14 m/s
- B: $(80\hat{i} + 100\hat{j})$ m, 10 m/s
- C: $(100\hat{i} + 100\hat{j})$ m, 14 m/s
- D: $(50\hat{i} + 150\hat{j})$ m, 10 m/s

Review of Geometry and Trigonometry Techniques

Review of Geometry and Trigonometry Techniques

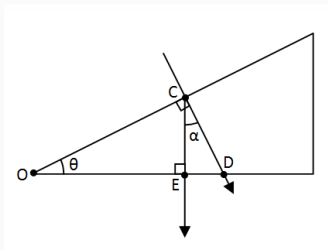


Figure 10: Angles of a triangle add up to π (180°).

$$\angle OCD = \pi/2 = \angle OCE + \angle DCE = \angle OCE + \alpha \quad (4)$$

$$\angle OCE + \angle COE + \pi/2 = \pi = \angle OCE + \theta + \pi/2 \quad (5)$$

$$\angle OCE + \theta = \pi/2 \quad (6)$$

$$\theta = \alpha \quad (7)$$

Review of Geometry and Trigonometry Techniques

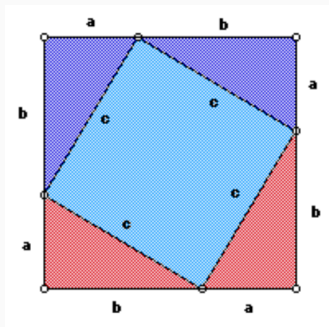


Figure 11: Proof of Pythagorean theorem.

$$A_1 = (a + b)^2 = a^2 + b^2 + 4((1/2)ab) \quad (8)$$

$$A_2 = c^2 + 4((1/2)ab) = A_1 \quad (9)$$

$$a^2 + b^2 = c^2 \quad (10)$$

Review of Geometry and Trigonometry Techniques

One soccer teammate passes the ball to another. The player without the ball is 7 meters away from the player with the ball, and they are both running in the same direction. The player without the ball runs ahead by 24 meters before the pass. How far does the ball travel?

- A: 7 meters
- B: 24 meters
- C: 25 meters
- D: 17 meters

Review of Geometry and Trigonometry Techniques

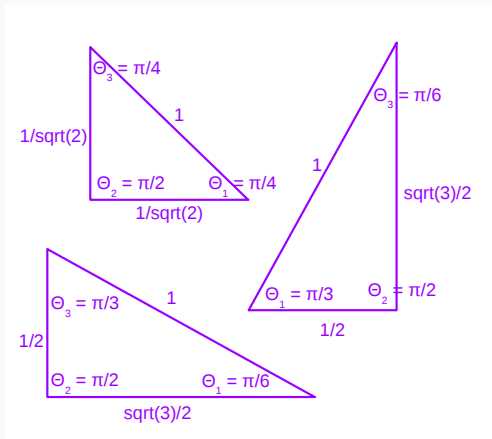


Figure 12: Memorize the properties of these special triangles.

Review of Geometry and Trigonometry Techniques

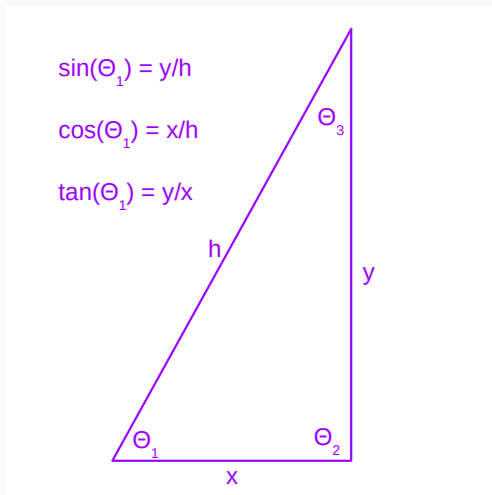


Figure 13: Working definitions of trigonometric functions.

Review of Geometry and Trigonometry Techniques

What is $\sin(30^\circ)$?

- A: $1/2$
- B: $\sqrt{3}/2$
- C: 0
- D: 1

What is $\tan(45^\circ)$?

- A: $1/2$
- B: $\sqrt{3}/2$
- C: 0
- D: 1

Review of Geometry and Trigonometry Techniques

What is $\sin(30^\circ)^2 + \cos(30^\circ)^2$?

- A: $1/2$
- B: $\sqrt{3}/2$
- C: 0
- D: 1

A right-triangle has sides of length 3, 4, and a hypotenuse of 5. What are the angles inside the triangle?

- A: $\arctan(5/4)$, $\arctan(4/5)$, $\pi/2$
- B: $\arctan(1)$, $\arctan(4/5)$, $\pi/2$
- C: $\arctan(4/3)$, $\arctan(3/4)$, $\pi/2$
- D: $\arctan(3/4)$, $\arctan(3/4)$, $\pi/2$

Conclusion

Week 1 Summary

Chapters 1, 2.

1. Methods of approximation
 - **Estimating** the correct order of magnitude
 - **Function** approximation
 - **Unit analysis**
2. Coordinates and vectors
 - **Scalars** and **vectors**
 - **Cartesian** (rectangular) coordinates, displacement
 - **Vector** addition, subtraction, and multiplication
3. Review of geometry and trigonometry techniques
 - Parallel lines, similar triangles
 - Pythagorean theorem
 - Sine, cosine, tangent ...